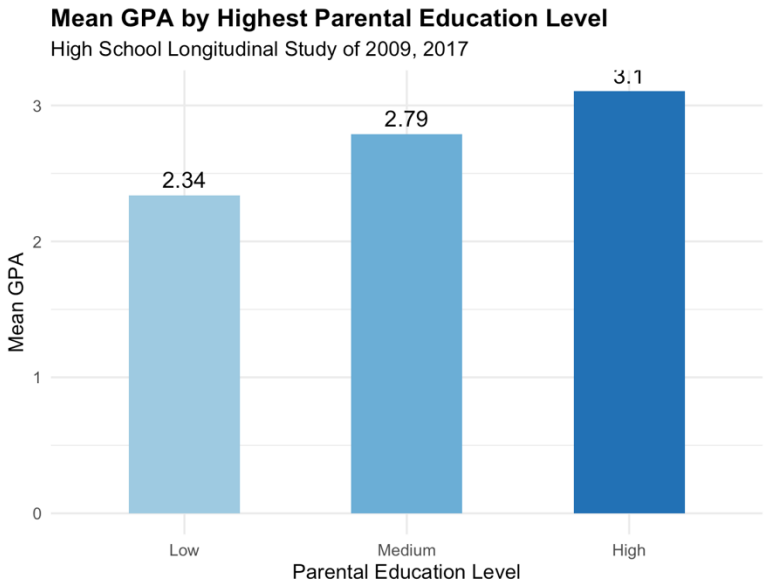
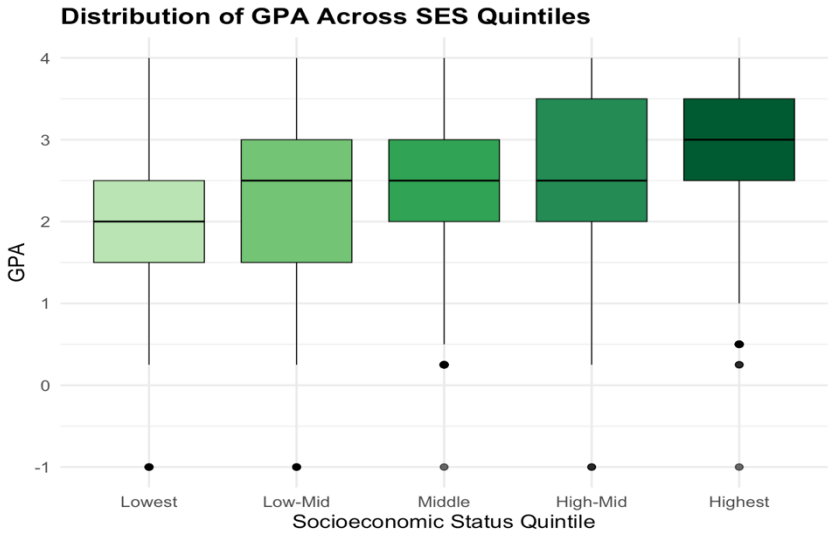
How Parental Education Shapes Children's Academic Achievement Across SES and Race

Background: Parental education strongly predicts children's academic success, yet socioeconomic (**SES**) and racial disparities persist in the US. Understanding family background's impact on achievement informs educational equity and policy, highlighting structural inequalities and identifying groups needing targeted support.

1. Data Collection Methodology: This study uses data from the High School Longitudinal Study of 2009 (**HSLS:09**), collected by NCES through student, parent, and school surveys. Because the study follows students over time, some missing data occur due to nonresponse and attrition. Several variables are self-reported, and smaller racial groups have limited sample sizes, restricting subgroup comparisons.



2. Variables Used, Measurement and Coding: Parental education - originally measured separately for each parent, was combined into a single "**highest parental education**" variable (Low, Medium, High). Student achievement was measured by **GPA** (0–4). **SES** was recoded from NCES SES quintiles, and race was simplified using NCES codes.

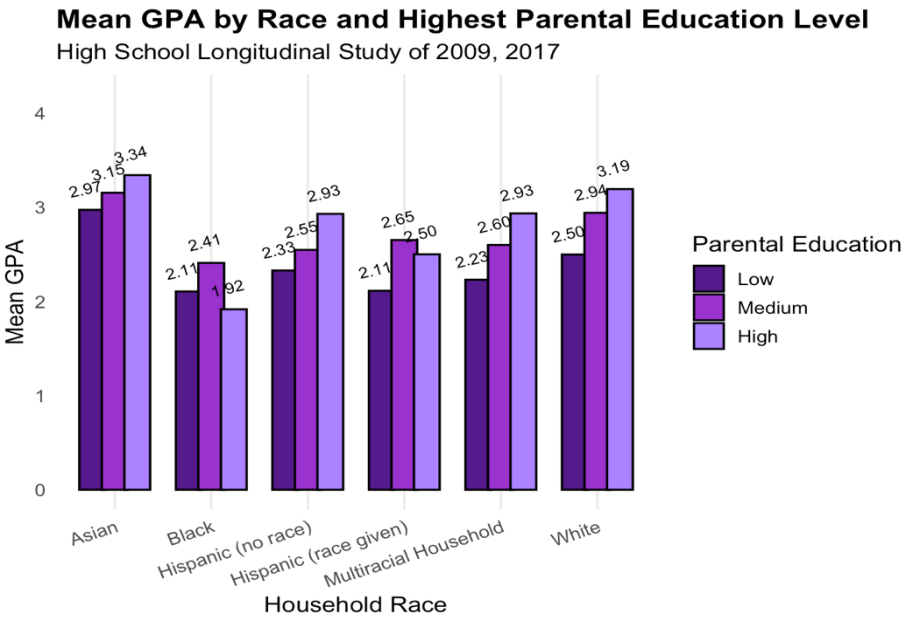


3. Analysis Approach: The analysis examined **GPA** differences across parental education levels, SES, and race using descriptive statistics. All missing values were recoded to NA before analysis. Subgroup means were computed using **dplyr**, and visualizations were created in **ggplot2**, including bar charts, boxplots, and a faceted chart.

4. Results: Students with higher-educated parents demonstrated higher **mean GPAs**. GPA also increased across SES quintiles, indicating a strong socioeconomic gradient in academic outcomes. Racial differences in GPA remained even within the same parental education levels, with White and Asian students generally scoring higher than other groups.

5. Conclusion: Higher parental education is associated with **higher academic performance** among students, even after considering differences in SES and race, pointing to inequalities that extend beyond individual effort or school factors.

6. Limitations and Future Directions: The study is limited by missing data, especially for smaller racial groups and students whose parents did not respond. Because the analysis is **descriptive**, it cannot determine whether parental education causes higher GPA. Future research using multivariate or longitudinal models and more complete data on underrepresented groups would improve the accuracy of subgroup comparisons.



References: National Center for Education Statistics. (2009). *High School Longitudinal Study of 2009 (HSLS:09) [Data file]*. U.S. Department of Education. https://nces.ed.gov/surveys/hsls09/